INCH-POUND

MIL-DTL-27426/3
22 SEPTEMBER 1997

DETAIL SPECIFICATION

RINGS, RETAINING, SPIRAL
(UNIFORM CROSS SECTION)
MEDIUM DUTY INTERNAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-27426, listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

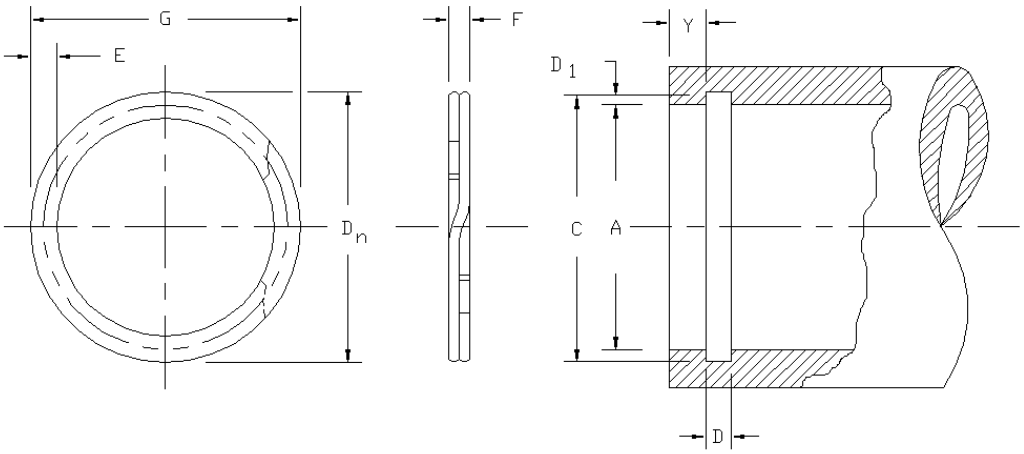


FIGURE 1 INTERNAL RING

Maximum bottom groove radius is 0.005 inches for rings up to 1.063 inches free diameter and 0.010 inches for larger rings. Ring thickness applies only to unplated rings. Add 0.002 inches to upper thickness tolerance for plated rings. Groove widths listed will accept rings without modification.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to : the Defense Industrial Supply Center, (Code DISC-EED), 700 Robbins Avenue, Philadelphia, PA 19111-5096, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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TABLE I. MEDIUM DUTY INTERNAL RETAINING RINGS.

DASH NO.	SHAFT DIAMETER		RING DIMENSIONS						GROOVE DIMENSIONS				
	INCHES A	MM	FREE DIAMETER G	TOL	THICKNESS F	TOL	RADIAL WALL E	TOL	DIAMETER C	TOL	WIDTH D	TOL	NOMINAL GROOVE DEPTH D_1
0100	.500	12.7	.532		.025		.045		.526		.030		.014
0101	.512	13.0	.544		.025		.045		.538		.030		.014
0102	.531	13.5	.564		.025		.045		.557		.030		.014
0103	.562	14.3	.594		.025		.045		.588		.030		.014
0104	.594	15.1	.626		.025		.045		.619	±.002	.030		.014
0105	.625	15.9	.658		.025		.045		.651		.030		.014
0106	.656	16.7	.689		.025		.045		.682		.030		.014
0107	.687	17.5	.720		.025		.045		.713		.030		.014
0108	.718	18.3	.751		.025		.045		.744		.030		.014
0109	.750	19.0	.790		.031		.065		.782		.036		.017
0110	.777	19.7	.817	+.013	.031		.065		.808		.036		.017
0111	.781	19.8	.821	−.000	.031		.065		.812		.036		.017
0112	.812	20.6	.853		.031		.065		.843		.036	+.003	.017
0113	.843	21.4	.889		.031		.065		.880		.036	−.000	.017
0114	.866	22.0	.913		.031		.065		.903		.036		.020
0115	.875	22.2	.922		.031		.065		.912	±.003	.036		.020
0116	.906	23.0	.949		.031		.065		.939		.036		.020
0117	.938	23.8	.986		.031		.065		.975		.036		.020
0118	.968	24.4	1.025		.037		.075		1.015		.042		.023
0119	.987	25.0	1.041		.037		.075		1.030		.042		.023
0120	1.000	25.4	1.054		.037	±.002	.075		1.043		.042		.023
0121	1.023	26.0	1.078		.037		.075		1.066		.042		.023
0122	1.031	26.3	1.084		.037		.075		1.074		.042		.023
0123	1.062	27.0	1.117		.037		.075	+.003	1.104		.042		.023
0124	1.093	27.8	1.147		.037		.075	−.005	1.135		.042		.023
0125	1.125	28.6	1.180		.037		.075		1.167		.042		.023
0126	1.156	29.3	1.210		.037		.075		1.198		.042		.023
0127	1.188	30.2	1.249		.043		.085		1.236		.048		.026
0128	1.218	31.0	1.278		.043		.085		1.266		.048		.026
0129	1.250	31.7	1.312		.043		.085		1.298	±.004	.048		.026
0130	1.281	32.6	1.342	−.000	.043		.085		1.329		.048		.026
0131	1.312	33.3	1.374	+.015	.043		.085		1.360		.048		.026
0132	1.343	34.1	1.408		.043		.085		1.395		.048		.026
0133	1.375	34.9	1.442		.043		.095		1.427		.048		.028
0134	1.406	35.8	1.472		.043		.095		1.458		.048		.028
0135	1.437	36.5	1.504		.043		.095		1.489		.048		.028
0136	1.456	37.0	1.523		.043		.095		1.508		.048		.028
0137	1.468	37.3	1.535		.043		.095		1.520		.048		.028
0138	1.500	38.1	1.567		.043		.095		1.552		.048		.028
0139	1.562	39.6	1.634		.049		.108		1.617		.056		.032
0140	1.574	40.0	1.649		.049		.108		1.633		.056		.032
0141	1.625	41.7	1.701		.049		.108		1.684		.056		.032
0142	1.653	42.0	1.730		.049		.108		1.712		.056	+.004	.032
0143	1.687	42.8	1.768		.049		.118		1.750		.056	−.000	.034
0144	1.750	44.4	1.834	−.000	.049		.118		1.813		.056		.034
0145	1.813	46.0	1.894	+.020	.049		.118		1.875		.056		.034
0146	1.850	47.0	1.937		.049	±.003	.118		1.917		.056		.036
0147	1.875	47.6	1.960		.049		.118		1.942		.056		.036
0148	1.938	49.2	2.025		.049		.118		2.005		.056		.036
0149	2.000	50.8	2.091		.049		.128		2.071		.056		.038
0150	2.047	52.0	2.138		.049		.128		2.118	±.005	.056		.038
0151	2.062	52.3	2.154	−.000	.049		.128		2.132		.056		.038
0152	2.125	53.9	2.217	+.025	.049		.128		2.195		.056		.038
0153	2.165	55.0	2.260		.049		.138		2.239		.056		.040

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TABLE I. MEDIUM DUTY INTERNAL RETAINING RINGS. (CONT'D)

DASH NO.	SHAFT DIAMETER		RING DIMENSIONS						GROOVE DIMENSIONS				
	INCHES A	MM	FREE DIAMETER G	TOL	THICKNESS F	TOL	RADIAL WALL E	TOL	DIAMETER C	TOL	WIDTH D	TOL	NOMINAL GROOVE DEPTH D_1
0154	2.188	55.5	2.284		.049		.138		2.262		.056		.040
0155	2.250	57.1	2.347		.049		.138		2.324		.056		.040
0156	2.312	58.7	2.413		.049		.138		2.390		.056		.042
0157	2.375	60.3	2.476		.049		.138		2.453	±.006	.056		.042
0158	2.437	61.9	2.543		.049		.148		2.519		.056		.044
0159	2.440	62.0	2.546		.049		.148		2.522		.056		.044
0160	2.500	63.5	2.606		.049		.148		2.582		.056	+.004	.044
0161	2.531	64.2	2.641		.049		.148		2.617		.056	−.000	.046
0162	2.562	65.1	2.673		.049		.148		2.648		.056		.046
0163	2.625	66.6	2.736	−.000	.049		.148		2.711		.056		.046
0164	2.677	68.0	2.789	+.025	.049	+.003	.158		2.767		.056		.048
0165	2.688	68.2	2.803		.049		.158		2.778		.056		.048
0166	2.750	69.8	2.865		.049		.158		2.841		.056		.048
0167	2.813	71.4	2.929		.049		.158		2.903		.056		.048
0168	2.834	71.9	2.954		.049		.168		2.928		.056		.050
0169	2.875	73.0	2.995		.049		.168		2.969		.056		.050
0170	2.937	74.5	3.058		.049		.168		3.031		.056		.050
0171	2.952	75.0	3.073		.049		.168		3.046		.056		.050
0172	3.000	76.1	3.122		.061		.168		3.096		.068		.051
0173	3.062	77.7	3.186		.061		.168		3.158		.068		.051
0174	3.125	79.3	3.251		.061		.178		3.223		.068		.052
0175	3.149	80.0	3.276		.061		.178		3.247		.068		.052
0176	3.187	81.0	3.311		.061		.178		3.283		.068		.052
0177	3.250	82.5	3.379		.061		.178		3.350		.068		.053
0178	3.312	84.4	3.446		.061		.188		3.416		.068		.055
0179	3.346	84.9	3.479		.061		.188		3.450		.068		.055
0180	3.375	85.7	3.509		.061		.188		3.479		.068		.055
0181	3.437	87.2	3.574		.061		.188		3.543		.068		.056
0182	3.500	88.8	3.636	+.030	.061		.188		3.606		.068		.056
0183	3.543	89.9	3.684	−.000	.061		.198		3.653		.068		.058
0184	3.562	90.5	3.703		.061		.198		3.672	±.006	.068		.058
0185	3.625	92.0	3.769		.061		.198		3.737		.068		.059
0186	3.687	93.6	3.832		.061		.198		3.799		.068		.059
0187	3.740	95.0	3.885		.061	±.003	.198		3.852		.068		.059
0188	3.750	95.2	3.894		.061		.198		3.862		.068	+.005	.059
0189	3.812	96.8	3.963		.061		.208		3.930		.068	−.000	.062
0190	3.875	98.3	4.025		.061		.208		3.993		.068		.062
0191	3.938	99.9	4.089		.061		.208		4.056		.068		.062
0192	4.000	101.5	4.157		.061		.218		4.124		.068		.065
0193	4.063	103.2	4.222		.061		.218		4.187		.068		.065
0194	4.125	104.8	4.284		.061		.218	+.004	4.249		.068		.065
0195	4.188	106.5	4.347		.061		.218	−.006	4.311		.068		.065
0196	4.250	107.9	4.416		.061		.228		4.380		.068		.068
0197	4.312	109.7	4.479		.061		.228		4.442		.068		.068
0198	4.330	109.9	4.497		.061		.228		4.460		.068		.068
0199	4.375	111.0	4.543		.061		.228		4.505		.068		.068
0200	4.437	112.7	4.611		.061		.238		4.573		.068		.071
0201	4.500	114.2	4.674	−.000	.061		.238		4.636		.068		.071
0202	4.527	115.0	4.701	+.035	.061		.238		4.663		.068		.071
0203	4.562	116.0	4.737		.061		.238		4.698		.068		.071
0204	4.625	117.4	4.803		.072		.250		4.765		.079		.073
0205	4.687	119.0	4.867		.072		.250		4.827		.079		.073
0206	4.724	120.0	4.903		.072	±.004	.250		4.864		.079		.073
0207	4.750	120.6	4.930		.072		.250		4.890		.079		.073
0208	4.812	122.2	4.993		.072		.250		4.952		.079		.073
0209	4.875	123.8	5.055		.072		.250		5.015		.079		.073
0210	4.921	125.0	5.102		.072		.250		5.061		.079		.073
0211	4.937	125.4	5.122		.072		.250		5.081		.079		.075

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TABLE I. MEDIUM DUTY INTERNAL RETAINING RINGS. (CONT'D)

DASH NO.	SHAFT DIAMETER		RING DIMENSIONS						GROOVE DIMENSIONS				
	INCHES A	MM	FREE DIAMETER G		THICKNESS F		RADIAL WALL E		DIAMETER C		WIDTH D		NOMINAL GROOVE DEPTH D_1
				TOL		TOL		TOL		TOL		TOL	
0212	5.000	126.9	5.185	-.000	.072		.250		5.144	±.006	.079		.075
0213	5.118	130.0	5.304	+.035	.072		.250		5.262		.079		.075
0214	5.125	130.1	5.311		.072		.250		5.269		.079		.075
0215	5.250	133.2	5.436		.072		.250		5.393		.079		.075
0216	5.375	136.5	5.566		.072		.250	+.004	5.522		.079	+.005	.077
0217	5.500	139.7	5.693		.072		.250	-.006	5.647		.079	-.000	.077
0218	5.511	140.0	5.703		.072		.250		5.658		.079		.077
0219	5.625	142.8	5.818	+.045	.072		.250		5.772	±.007	.079		.077
0220	5.708	145.0	5.909	-.000	.072		.250		5.861		.079		.080
0221	5.750	145.9	5.950		.072		.250		5.903		.079		.080
0222	5.875	149.2	6.077		.072		.250		6.028		.079		.080
0223	5.905	150.0	6.106		.072		.250		6.058		.079		.080
0224	6.000	152.3	6.202		.072		.250		6.153		.079		.080
0225	6.125	155.6	6.349		.086		.312		6.297		.094		.090
0226	6.250	158.8	6.474		.086		.312		6.422		.094		.090
0227	6.299	160.0	5.524		.086		.312		6.471		.094		.090
0228	6.375	162.0	6.601		.086		.312		6.547		.094		.090
0229	6.500	165.1	5.726		.086	±.004	.312		6.672		.094		.090
0230	6.625	168.3	6.863	+.055	.086		.312		6.807		.094		.095
0231	6.692	170.0	6.931	-.000	.086		.312		6.874		.094		.095
0232	6.750	171.4	5.987		.086		.312		6.932		.094		.095
0233	6.875	174.6	7.114		.086		.312		7.057		.094		.095
0234	7.000	177.8	7.239		.086		.312		7.182		.094		.095
0235	7.086	180.0	7.337		.086		.312		7.278		.094		.100
0236	7.125	181.0	7.376		.086		.312		7.317		.094		.100
0237	7.250	184.1	7.501		.086		.312		7.442		.094		.100
0238	7.375	187.3	7.628		.086		.312		7.567		.094		.100
0239	7.480	190.0	7.734		.086		.312	+.004	7.672		.094		.100
0240	7.500	190.5	7.754		.086		.312	-.008	7.692	±.008	.094		.100
0241	7.625	193.7	7.890		.086		.312		7.827		.094	+.006	.105
0242	7.750	196.8	8.014		.086		.312		7.952		.094	-.000	.105
0243	7.875	200.0	8.131		.086		.312		8.077		.094		.105
0244	8.000	203.0	8.266		.086		.375		8.202		.094		.105
0245	8.250	209.5	8.528		.086		.375		8.462		.094		.110
0246	8.267	210.0	8.546		.086		.375		8.479		.094		.110
0247	8.464	215.0	8.744		.086		.375		8.676		.094		.110
0248	8.500	216.0	8.780		.086		.375		8.712		.094		.110
0249	8.750	222.2	9.041	+.065	.086		.375		8.972		.094		.115
0250	8.858	225.0	9.151	-.000	.086	±.004	.375		9.080		.094		.115
0251	9.000	228.6	9.293		.086		.375		9.222		.094		.115
0252	9.055	230.0	9.359		.086		.375		9.287		.094		.120
0253	9.250	235.0	9.555		.086		.375		9.482		.094		.120
0254	9.448	240.0	9.755		.086		.375		9.680		.094		.120
0255	9.500	241.3	9.806		.086		.375		9.732		.094		.120
0256	9.750	247.8	10.068		.086		.375		9.992		.094		.125
0257	10.000	254.0	10.320		.086		.375		10.242		.094		.125
0258	10.250	260.4	10.582		.086		.375		10.502		.094		.130
0259	10.500	265.0	10.834		.086		.375		10.752		.094		.130
0260	10.750	273.0	11.095		.086		.375		11.012		.094		.135
0261	11.000	279.4	11.347		.086		.375		11.262		.094		.135

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